```
str_1 = "Use \t this formula when \n your data includes \\ every single item or person you want to \'study\'"
print(str_1)
```

```
Use      this formula when
 your data includes \ every single item or person you want to 'study'
```

```
zaheela = 65
sarah = 75
atif = 85
abdullah = 90
print(zaheela)
print(abdullah)
```

```
65
90
```

```
#list: To store multiple data in one variable.List are mutable(can be change)
#list used square brakets [] #dictionary
marks = [65,75,85,90]
marks
```

```
[65, 75, 85, 90]
```

```
bio_data = ["uzair","shakir",35,3.9,True,None] #Datatypes(5 types)
#int,str,float,boolean,none
bio_data
```

```
['uzair', 'shakir', 35, 3.9, True, None]
```

```
type(bio_data)
```

```
list
```

```
len(bio_data)
```

```
6
```

```
#Indexing #T = 0 , h = 1, i = 2, s = 3, space = 4......
str_2 = "This is python"
str_2[0]
```

```
'T'
```

```
student = ["Osama","hassan",3.9,"A",20,True]
student[3]
```

```
'A'
```

```
student = ["Osama","hassan",3.9,"A",20,True," "] #string = " "
student[6]
```

```
' '
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
print(student)
```

```
['Osama', 'hassan', 3.9, 'A', 20, True, ' ']
```

```
a = 39
type(a)
```

```
int
```

```
a = "39"
type(a)
```

```
str
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
student[7]
```

```
---------------------------------------------------------------------------
IndexError                                Traceback (most recent call last)
/tmp/ipykernel_453/1810518954.py in <cell line: 0>()
      1 student = ["Osama","hassan",3.9,"A",20,True," "]
----> 2 student[7]

IndexError: list index out of range
```

Next steps: Explain error

```
student
```

```
['Osama', 'hassan', 3.9, 'A', 20, True, ' ']
```

```
student = ["Osama","hassan",3.9,"A",20,True," "]
student[2] = 3.5
student
```

```
['Osama', 'hassan', 3.5, 'A', 20, True, ' ']
```

```
bio_data = ["ali","Truck Driver"]
bio_data = ["muhammad","Truck","Driver"]
```

```
#slicing
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[0:2] #[startind_index,ending_index(does not count)]
```

```
['ali', 'Truck Driver']
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[1:]
```

```
['Truck Driver', 42, 'matric', 4.5]
```

```
bio_data[:2]
```

```
['ali', 'Truck Driver']
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[-1]
bio_data[-2]
```

```
'matric'
```

```
bio_data[-3:-1]
```

```
[42, 'matric']
```

```
bio_data[-3:]
```

```
[42, 'matric', 4.5]
```

```
bio_data[-1:]#[lower_number:highest_number]
```

```
[4.5]
```

```
bio_data[-1:-3]
```

```
[]
```

```
bio_data = ["ali","Truck Driver",42,"matric",4.5]
bio_data[1:3]
```

```
['Truck Driver', 42]
```

```
#list_methods
num = [1,2,3,4,5,6,7]
num.append(8)
num
```

```
[1, 2, 3, 4, 5, 6, 7, 8]
```

```
integer = [2,1,4,5,7,6,3]
integer.sort()
integer#ascending order
```

```
[1, 2, 3, 4, 5, 6, 7]
```

```
integer.sort(reverse=True)
integer #descending order
```

```
[7, 6, 5, 4, 3, 2, 1]
```

```
integer = [2,1,4,5,7,6,3]
integer.reverse()
integer
```

```
[3, 6, 7, 5, 4, 1, 2]
```

```
fruits = ["grapes","kiwi","oranges","apple","pineapple","leechi"]
fruits.sort()
fruits
```

```
['apple', 'grapes', 'kiwi', 'leechi', 'oranges', 'pineapple']
```

```
number = [1,5,9] #insert(index,element)
number.insert(2,4)
number
```

```
[1, 5, 4, 9]
```

```
cls = [1,2,1,3,4,1] #remove(element)
cls.remove(1)
cls
```

```
[2, 1, 3, 4, 1]
```

```
name = ["uzair","shakir",1,"uzair"]
name.remove("uzair") #First occurance
name
```

```
['shakir', 1, 'uzair']
```

```
name = ["uzair","shakir",1,"uzair"]
name.pop(1) #pop(index_number)
name
```

```
['uzair', 1, 'uzair']
```

```
a = [1,2,3,4]
b = a.copy()
b.reverse()
b
```

```
[4, 3, 2, 1]
```

```
c = [3,1,2,6,5]
d = c.copy()
d.sort()
d
```

```
[1, 2, 3, 5, 6]
```

```
name = ["uzair","shakir",1,"uzair"]
name.pop(1) #pop(index_number)
name
```

```
['uzair', 1, 'uzair']
```

```
name = ["uzair","shakir",1,"uzair"]
name.count("uzair")
```

```
2
```

```
Start coding or generate with AI.
```